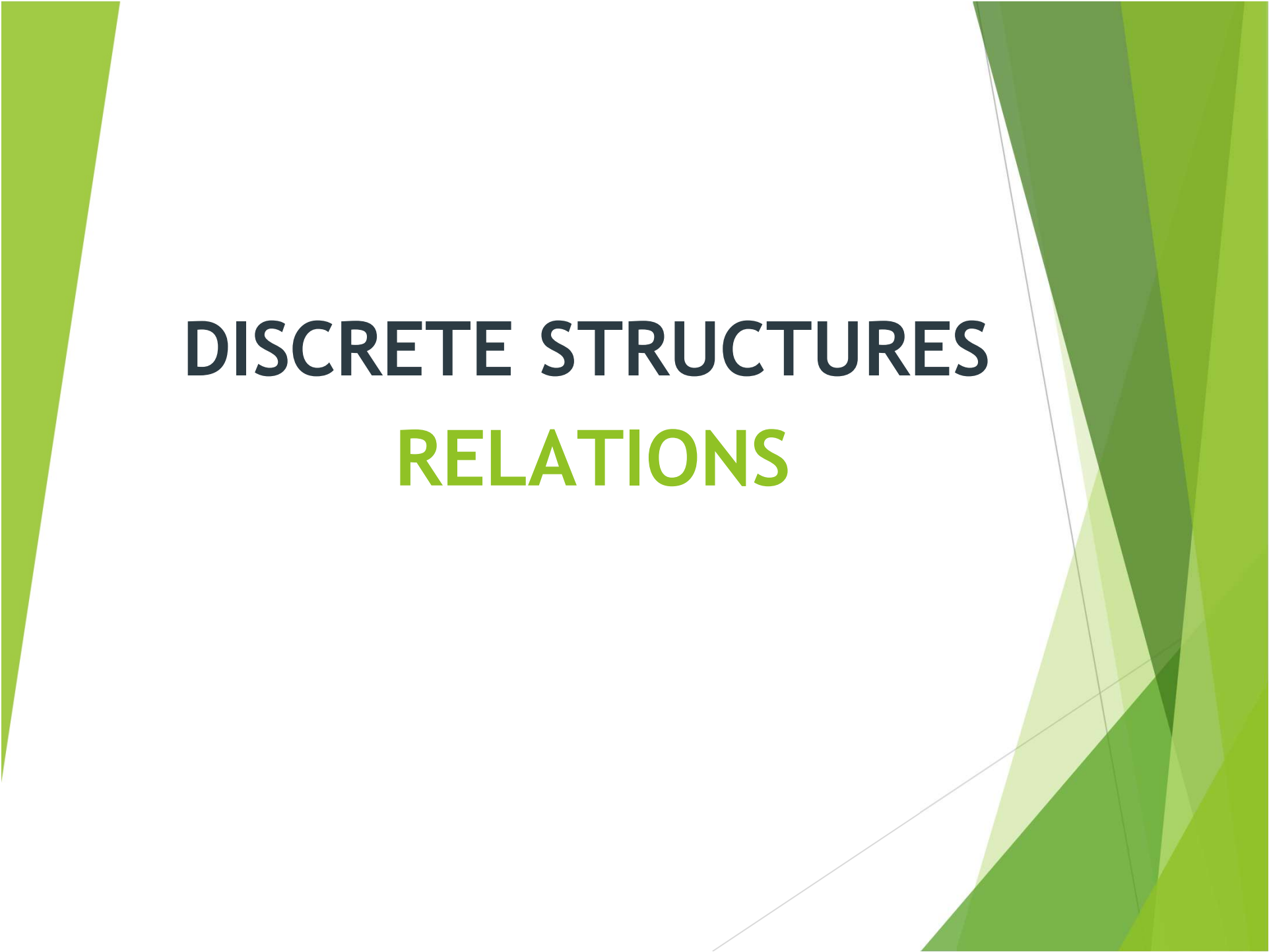


The background features abstract green geometric shapes. On the left, a single green triangle points downwards. On the right, a complex arrangement of overlapping green triangles and polygons in various shades of green (from light lime to dark forest green) creates a layered, architectural effect. A thin, light gray line extends diagonally from the bottom left towards the right side of the composition.

DISCRETE STRUCTURES

RELATIONS

WHY STUDY RELATIONS ?

We study relations because they help us understand and model **connections between things** in mathematics, computer science, and real life.

▶ In Mathematics:

Relations help us study connections between numbers.

For example: One number **divides** another (like 2 divides 6).

▶ In Computers:

Relations describe connections such as:

- A **program** and the **variables** it uses.
- A **programming language** and its **valid statements**.
- A **database** storing related information.

▶ In Real-World Problems:

Relations help us solve practical problems, such as:

- Finding which **cities are connected by flights**.
- Deciding the correct **order of tasks in a project**.

STRUCTURES FOR REPRESENTING RELATIONS

- Relationships between **elements of two sets** are represented using the structure called a **binary relation**, which is just a subset of the Cartesian product of the Sets.

Example: "Student studies Subject", $A = \{\text{Ali, Ahmed}\}$

$B = \{\text{Math, English}\}$

$R = \{(\text{Ali, Math}), (\text{Ahmed, English})\}$ This is a binary relation because:

- It connects elements of two sets,
- Each pair has 2 elements
- $R \subseteq A \times B$

- Relationships between **elements of more than two sets arise in many contexts**. These relationships can be represented by **n -ary relations**, which are collections of n -tuples.

Example: (Student, Course, Grade),

$R = \{(\text{Ali, Math, 85}), (\text{Ahmed, English, 90})\}$ -R involve more than two sets.

- Such relations are the basis of the relational data model, the most common way to store information in computer databases.

METHODS FOR REPRESENTING RELATIONS

► Two methods for representing relations:

1. **Square Matrices**
2. **Directed Graphs** (consisting of vertices and directed edges)

BINARY RELATIONS AND THEIR PROPERTIES

- ▶ The easiest way to show a relationship between two sets is by using ordered pairs like (a, b) .
- Because we use pairs of two elements, this is called a binary relation.

DEFINITION: Let A and B be sets. A *binary relation from A to B* is a subset of $A \times B$.

- ▶ In other words, a binary relation from A to B is a set R of ordered pairs, where the first element of each ordered pair comes from A and the second element comes from B .
- ▶ We use the notation $a R b$ to denote that $(a, b) \in R$ and $a \not R b$ to denote that $(a, b) \notin R$.
- ▶ Moreover, If (a, b) belongs to R , we say: “ a is related to b by R ”.
- ▶ Binary relations represent relationships between the elements of two sets

ORDERED PAIR & CARTESIAN PRODUCT

- ▶ The definition of a set explicitly disregards the order of the set elements, However, sometimes the order is important.
- ▶ This leads to the notion of **an ordered pair** of two elements x and y , denoted (x, y) .
- ▶ The crucial property is:

(x, y) EQUALS TO (u, v) if and only if $x = u$ and $y = v$

- ▶ If the order changes, the pair is different. Example: $(1, 2) \neq (2, 1)$
- This notion can be extended naturally to define an ordered n -tuple as the ordered counterpart of a set with n elements.
- ▶ Give two sets A and B , their **cartesian product** $A \times B$ is the set of all ordered pairs (x, y) , such that $x \in A$ and $y \in B$:

$$A \times B = \{(x, y) : x \in A, y \in B\}$$

If $A = \{1, 2\}$, $B = \{a, b, c\}$ Then: $A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$

NOTE:

If set A has m elements and set B has n elements then $A \times B$ has $m \times n$ elements.

EXAMPLE A (ORDERED PAIR)

Find x and y given $(2x, x + y) = (6, 2)$

SOLUTION:

Two ordered pairs are equal if and only if the corresponding components are equal. Hence, we obtain the equations:

$$2x = 6 \quad \dots\dots\dots(1)$$

$$x + y = 2 \quad \dots\dots\dots(2)$$

Solving equation (1) we get $x = 3$ and when the value is substituted in (2) we get $y = -1$.

EXAMPLE B(CARTESIAN PRODUCT)

Let $A = \{1, 2\}$, $B = \{a, b, c\}$ then find $A \times B, B \times A, A \times A, B \times B$.

SOLUTION:

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$$

$$B \times A = \{(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)\}$$

$$A \times A = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$$

$$B \times B = \{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (c, c)\}$$

REMARK:

- ▶ $A \times B \neq B \times A$ for non-empty and unequal sets A and B .
- ▶ $A \times \phi = \phi \times A = \phi$
- ▶ $|A \times B| = |A| \times |B|$
- If $|A|=2$, $|B|=3$ so $|A \times B|=6$

EXAMPLE C (BINARY RELATION)

Let $A = \{1, 2\}$, $B = \{1, 2, 3\}$, List any five binary relations.

SOLUTION:

$$A \times B = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)\}$$

The Binary Relations over $A \times B$ are:

$$R1 = \{(1, 1), (1, 3), (2, 2)\}$$

$$R2 = \{(1, 2), (2, 1), (2, 2), (2, 3)\}$$

$$R3 = \{(1, 1)\}$$

$$R4 = A \times B$$

$$R5 = \emptyset$$

- All being subsets of $A \times B$ are relations from A to B .

DOMAIN & RANGE OF RELATIONS

DOMAIN OF A RELATION:

The domain of a relation R from A to B is the set of all first elements of the ordered pairs which belong to R denoted Dom(R).

Symbolically:

$$\text{Dom}(R) = \{a \in A \mid (a,b) \in R\}$$

RANGE OF A RELATION:

The range of A relation R from A to B is the set of all second elements of the ordered pairs which belong to R denoted Ran(R).

Symbolically:

$$\text{Ran}(R) = \{b \in B \mid (a,b) \in R\}$$

If: $R = \{(1,2), (1,3), (2,3)\}$

Domain: All first elements, **$\text{Dom}(R) = \{1,2\}$**

Range: All second elements, **$\text{Ran}(R) = \{2,3\}$**

EXAMPLE D

Let $A = \{1, 2\}$, $B = \{1, 2, 3\}$, Define a binary relation R from A to B as follows: $R = \{(a, b) \in A \times B \mid a < b\}$

Then

- a. Find the ordered pairs in R .
- b. Find the Domain and Range of R .
- c. Is $1R3$, $2R2$?

SOLUTION:

Given $A = \{1, 2\}$, $B = \{1, 2, 3\}$, then

$$A \times B = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3)\}$$

a) $R = \{(a, b) \in A \times B \mid a < b\}$

$$R = \{(1,2), (1,3), (2,3)\}$$

b) $\text{Dom}(R) = \{1,2\}$ and $\text{Ran}(R) = \{2, 3\}$

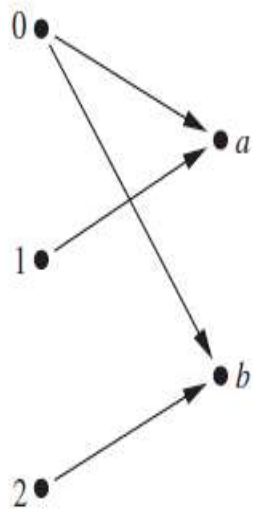
c) Since $(1,3) \in R$ so $1R3$, But $(2, 2) \notin R$ so 2 is not related.

EXAMPLE E

Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$.

Then $\{(0, a), (0, b), (1, a), (2, b)\}$ is a relation from A to B . This means, for instance, that $0Ra$, but that $1 \not R b$ & also $2 \not R a$.

Relations can be represented graphically, as shown in the figure below, using arrows to represent ordered pairs. Another way to represent this relation is to use a table, which is also shown in the figure.



R	a	b
0	×	×
1	×	
2		×

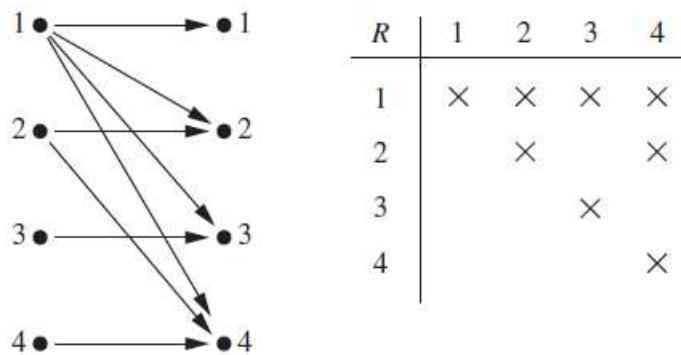
EXAMPLE F

Let A be the set $\{1, 2, 3, 4\}$. Which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$?

SOLUTION: Because (a, b) is in R if and only if a and b are positive integers not exceeding 4 such that a divides b , we see that

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}.$$

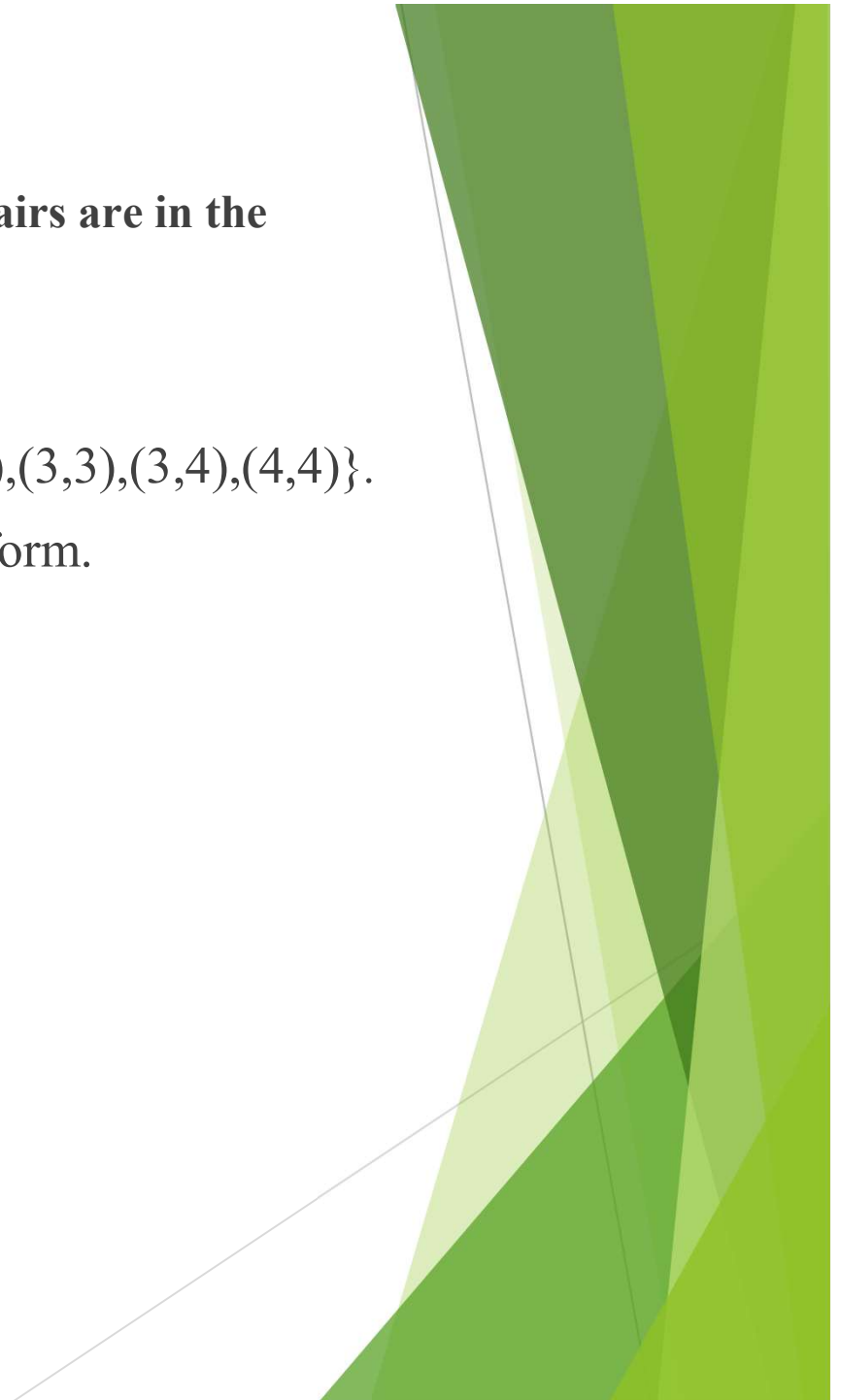
- The pairs in this relation are displayed both graphically and in tabular form in the figure.



Let A be the set $\{1, 2, 3, 4\}$. Which ordered pairs are in the relation $R = \{(a, b) \mid a \leq b\}$?

$R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$.

► represent R graphically and in tabular form.



EXERCISE A

1. Consider these relations on the set of integers:

$$R1 = \{(a, b) \mid a \leq b\},$$

$$R2 = \{(a, b) \mid a > b\},$$

$$R3 = \{(a, b) \mid a = b \text{ or } a = -b\},$$

$$R4 = \{(a, b) \mid a = b\},$$

$$R5 = \{(a, b) \mid a = b + 1\},$$

$$R6 = \{(a, b) \mid a + b \leq 3\}.$$

Which of these relations contain each of the pairs $(1, 1)$, $(1, 2)$, $(2, 1)$, $(1, -1)$, and $(2, 2)$?

2. List the ordered pairs in the relation R from $A = \{0, 1, 2, 3, 4\}$ to $B = \{0, 1, 2, 3\}$, where $(a, b) \in R$ if and only if

a) $a = b$.

b) $a + b = 4$.

c) $a > b$.

d) $a \mid b$.

e) $\gcd(a, b) = 1$.

f) $\text{lcm}(a, b) = 2$.

3. List all the ordered pairs in the relation $R = \{(a, b) \mid a \text{ divides } b\}$ on the set $\{1, 2, 3, 4, 5, 6\}$.

a) Display this relation graphically.

b) Display this relation in tabular form.

RELATIONS ON A SET

- Relations from a set A to itself are of special interest.

DEFINITION: A relation on a set A is a relation from A to A . In other words, a relation on a set A is a subset of $A \times A$.

EXAMPLE:

How many relations are there on a set with n elements?

SOLUTION:

A relation on a set A is a subset of $A \times A$.

Because $A \times A$ has n^2 elements when A has n elements, and a set with m elements has 2^m subsets, there are 2^{n^2} subsets of $A \times A$.

Thus, there are 2^{n^2} relations on a set with n elements.

For example, there are $2^{3^2} = 2^9 = 512$ relations on the set $\{a, b, c\}$.

RELATIONS ON A SET

Let:

$$A = \{1, 2\}$$

Then: $n = 2$

Step 1:

$A \times A$ has: $2^2 = 4$ elements

Those are:

$(1, 1), (1, 2), (2, 1), (2, 2)$ So $m = 4$

Step 2:

Number of subsets = $2^4 = 16$

So there are 16 relations.



PROPERTIES OF RELATIONS

There are several properties that are used to classify relations on a set.

A relation R on a set A is called

1. **reflexive** if for all $a \in A$, aRa .
Every element is related to itself.
2. **symmetric** if for all $a, b \in A$, aRb implies bRa .
If $(a,b) \in R$ then (b,a) must also be in R
3. **antisymmetric** if for all $a, b \in A$, aRb and bRa implies $a = b$.
If: $(a,b) \in R$ and $(b,a) \in R$ Then: $a = b$
4. **transitive** if for all $a, b, c \in A$, aRb and bRc implies aRc .
If: $(a,b) \in R$ and $(b,c) \in R$ Then: (a,c) must also be in R

REFLEXIVE RELATION

A relation R on a set A is called reflexive if $(a, a) \in R$ for every element $a \in A$.

$$(a,a) \in R$$

Every element is related to itself.

Example $A = \{1,2,3\}$

If $R = \{(1,1),(2,2),(3,3),(1,2)\}$

This is reflexive because:

$(1,1)$

$(2,2)$

$(3,3)$

All elements relate to themselves.

If one is missing?

If $(3,3)$ is missing $\rightarrow$ NOT reflexive.

REFLEXIVE RELATION

EXAMPLE G

Consider the following relations on $\{1, 2, 3, 4\}$:

- ▶ $R1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$,
- ▶ $R2 = \{(1, 1), (1, 2), (2, 1)\}$,
- ▶ $R3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$,
- ▶ $R4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$,
- ▶ $R5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$,
- ▶ $R6 = \{(3, 4)\}$.

Which of these relations are reflexive?

SOLUTION:

- ▶ The relations **R3 and R5** are **reflexive** because they both contain all pairs of the form (a, a) , namely, $(1, 1)$, $(2, 2)$, $(3, 3)$, and $(4, 4)$.
- ▶ The other relations are **not reflexive** because they do not contain all of these ordered pairs. In particular, **R1, R2, R4, and R6** are not reflexive because $(3, 3)$ is not in any of these relations.

SYMMETRIC & ANTISYMMETRIC RELATION

A relation R on a set A is called *symmetric* if $(b, a) \in R$ whenever $(a, b) \in R$, for all $a, b \in A$.

AND

A relation R on a set A such that for all $a, b \in A$, if $(a, b) \in R$ and $(b, a) \in R$, then $a = b$ is called *antisymmetric*.

- A **symmetric** relation:

A relation R on a set A is symmetric if: $(a,b) \in R \Rightarrow (b,a) \in R$

Meaning: Whenever a is related to b , b must also be related to a .

- Let $A = \{1, 2, 3\}$ so $R = \{(1, 2), (2, 1), (2, 3), (3, 2)\}$ So R is symmetric

- An **antisymmetric** relation:

A relation R on A is antisymmetric if: $(a,b) \in R$ and $(b,a) \in R \Rightarrow a = b$

Meaning: If both directions happen, then a and b must be the same element.

- Let $A = \{1, 2, 3\}$, $R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$

Notice: $(1, 2)$ is there $(2, 1)$ is NOT there

So no two different numbers point to each other.

EXAMPLE I

Consider the following relations on $\{1, 2, 3, 4\}$:

- ▶ $R2 = \{(1, 1), (1, 2), (2, 1)\}$,
- ▶ $R3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$,
- ▶ $R4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$,
- ▶ $R5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$,
- ▶ $R6 = \{(3, 4)\}$.

Which of these relations are Symmetric and antisymmetric ?

SOLUTION:

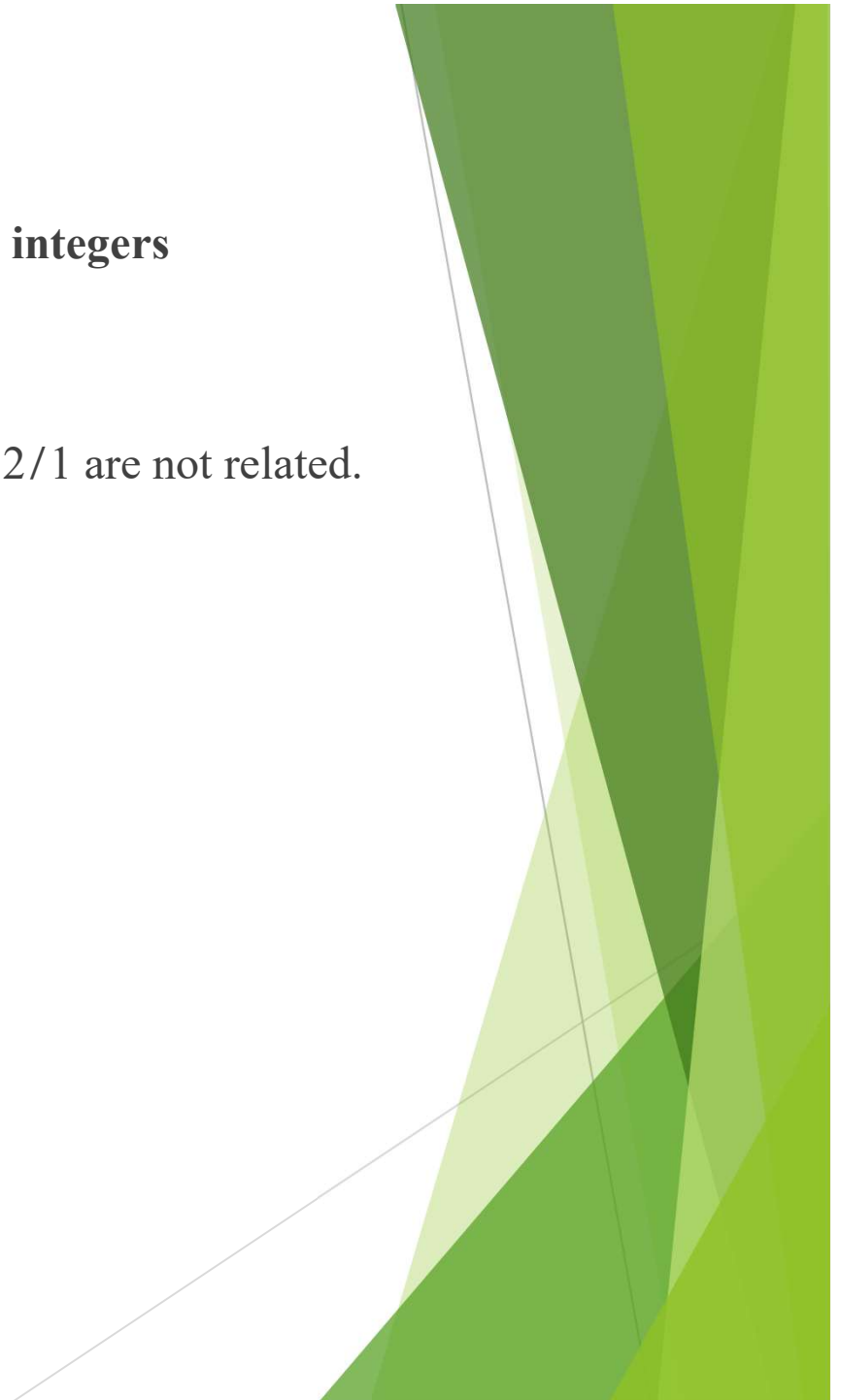
- ▶ **The relations $R2$ and $R3$ are symmetric**, because in each case (b, a) belongs to the relation whenever (a, b) does. For $R2$, the only thing to check is that both $(2, 1)$ and $(1, 2)$ are in the relation. For $R3$, it is necessary to check that both $(1, 2)$ and $(2, 1)$ belong to the relation, and $(1, 4)$ and $(4, 1)$ belong to the relation.
- ▶ **$R4$, $R5$, and $R6$ are all antisymmetric**. For each of these relations there is no pair of elements a and b with $a \neq b$ such that both (a, b) and (b, a) belong to the relation. This is done by finding a pair (a, b) with $a \neq b$ such that (a, b) and (b, a) are both in the relation.

EXAMPLE J

Is the “divides” relation on the set of positive integers symmetric?

SOLUTION:

This relation is not symmetric because $1/2$, are $2/1$ are not related.



TRANSITIVE RELATION

A relation R on a set A is called transitive if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, for all $a, b, c \in A$.

- If two-step paths exist, the shortcut must exist.
If no two-step path exists, nothing can go wrong.

NOTE:

Transitivity only requires that when a chain exists, the resulting pair (a, c) is in the relation.

If no chain exists, there's nothing to check $\rightarrow$ automatically OK

TRANSITIVE RELATION

A relation R on a set A is called transitive if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, for all $a, b, c \in A$.

EXAMPLE K

Consider the following relations on $\{1, 2, 3, 4\}$:

- ▶ $R1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$,
- ▶ $R2 = \{(1, 1), (1, 2), (2, 1)\}$,
- ▶ $R3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$,
- ▶ $R4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$,
- ▶ $R5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$.

Which of these relations are transitive ?

SOLUTION:

$R4$ and $R5$ are transitive. For each of these relations, we can show that it is transitive by verifying that if (a, b) and (b, c) belong to this relation, then (a, c) also does. For instance, $R4$ is transitive, because of $(3, 2)$ and $(2, 1)$, $(4, 2)$ and $(2, 1)$, $(4, 3)$ and $(3, 1)$, and $(4, 3)$ and $(3, 2)$ are the only such sets of pairs, and $(3, 1)$, $(4, 1)$, and $(4, 2)$ belong to $R4$.

$R1$ is not transitive because $(3, 4)$ and $(4, 1)$ belong to $R1$, but $(3, 1)$ does not.

$R2$ is not transitive because $(2, 1)$ and $(1, 2)$ belong to $R2$, but $(2, 2)$ does not.

$R3$ is not transitive because $(4, 1)$ and $(1, 2)$ belong to $R3$, but $(4, 2)$ does not.

Transitivity means: ALL possible chains must satisfy the rule.

EXERCISE B

- ▶ Is the “divides” relation on the set of positive integers transitive?
- ▶ For each of these relations on the set $\{1, 2, 3, 4\}$, decide whether it is reflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive.
 - a) $\{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$
 - b) $\{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$
 - c) $\{(2, 4), (4, 2)\}$
 - d) $\{(1, 2), (2, 3), (3, 4)\}$
 - e) $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$
 - f) $\{(1, 3), (1, 4), (2, 3), (2, 4), (3, 1), (3, 4)\}$
- ▶ Determine whether the relation R on the set of all people is reflexive, symmetric, antisymmetric, and/or transitive, where $(a, b) \in R$ if and only if
 - a) a is taller than b .
 - b) a and b were born on the same day.
 - c) a has the same first name as b .
 - d) a and b have a common grandparent.

Combining Relations

Given two relations R_1 and R_2 , we can combine them using basic set operations to form new relations such as $R_1 \cup R_2$, $R_1 \cap R_2$, $R_1 - R_2$, and $R_2 - R_1$.

Example: Let $A = \{1,2,3\}$ and $B = \{1,2,3,4\}$. The relations $R_1 = \{(1,1), (2,2), (3,3)\}$ and $R_2 = \{(1,1), (1,2), (1,3), (1,4)\}$ can be combined using basic set operations to form new relations:

Take all pairs that are in R1 OR in R2

$$R_1 \cup R_2 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (3,3)\}$$

Take only pairs that are in BOTH R1 AND R2

$$R_1 \cap R_2 = \{(1,1)\}$$

Take pairs that are in R1 but NOT in R2

$$R_1 - R_2 = \{(2,2), (3,3)\}$$

Take pairs that are in R2 but NOT in R1

$$R_2 - R_1 = \{(1,2), (1,3), (1,4)\}$$

Composition

Composition means **linking two relations through a common middle element**.

- Let R_1 connect set **A** to set **B**.
- Let R_2 connect set **B** to set **C**.

To form the composition $R_2 \circ R_1$ (read: “ R_2 after R_1 ”):

- If R_1 connects **x** to **y**,
 - And R_2 connects **y** to **z**,
 - Then the composition connects **x** to **z**.
- **Match the middle element (y), & connect the first to the last.**

Let the sets be:

- $A = \{1, 2\}$, $B = \{a, b\}$, $C = \{X, Y\}$
- $R_1 = \{ (1, a), (2, b) \}$
- $R_2 = \{ (a, X), (b, Y) \}$
- $(R_2 \circ R_1) = \{ (1, X), (2, Y) \}$

RELATIONS AND FUNCTIONS

A relation is just any set of ordered pairs like: (x, y)

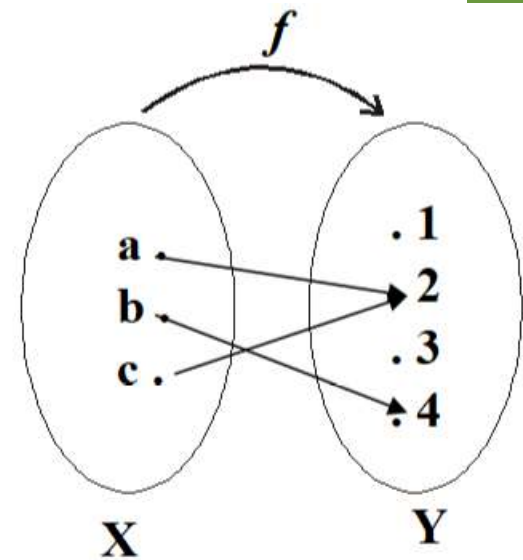
A function is a *special kind* of relation.

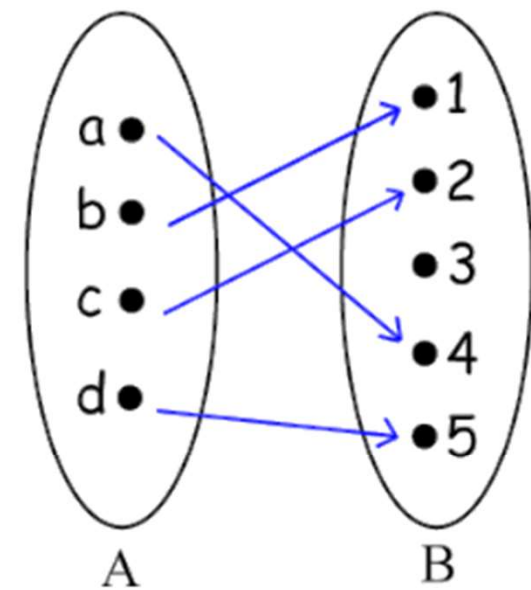
Since a function is made of ordered pairs too, **every function is automatically a relation.**

But not every relation is a function

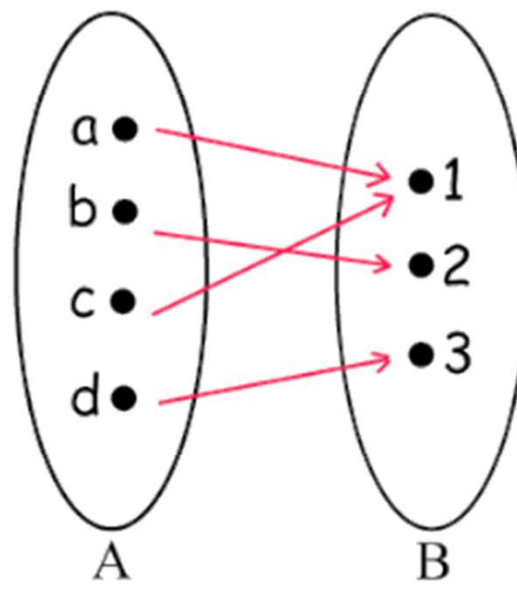
A function must follow **two strict rules**:

1. Every input must have an output.
2. Each input can have only one output.

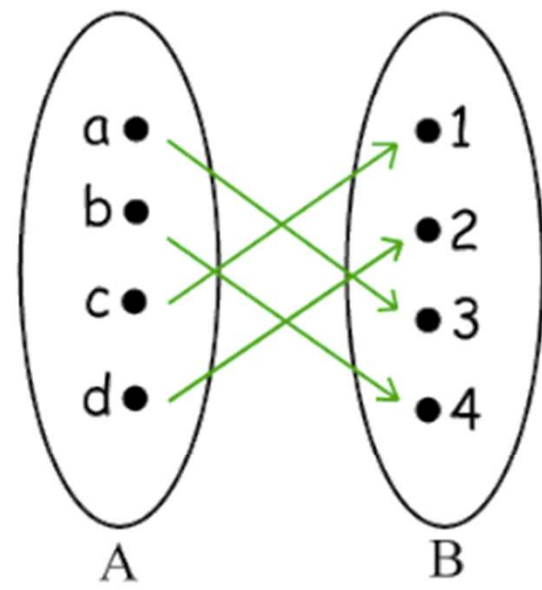




Injective
(one-to-one)

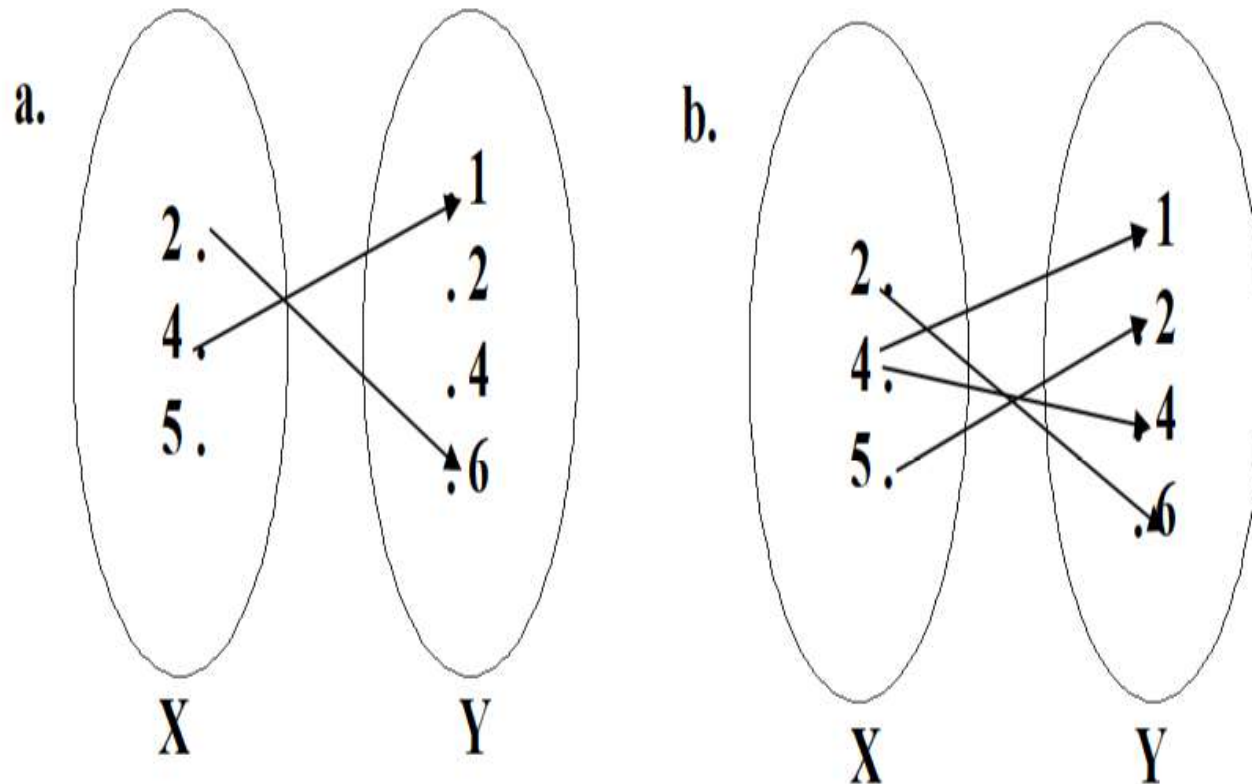


Surjective
(onto)



Bijjective
(one-to-one and onto)

FUNCTIONS & NONFUNCTIONS



The relation given in the diagram (a) is **Not a function** because there is no arrow coming out of $5 \in X$ to any element of Y.

The relation in the diagram (b) is **Not a function**, because there are two arrows coming out of $4 \in X$. i.e., $4 \in X$ is not related to a unique element of Y.

REPRESENTING RELATIONS

► There are many ways to represent a relation between finite sets.

► As we have seen previously, one way is to list its ordered pairs.

Another way to represent a relation is to use a table.

► Now we will discuss two alternative methods for representing relations. One method uses zero–one matrices. The other method uses pictorial representations called directed graphs.

REPRESENTING RELATIONS USING MATRICES

A relation between finite sets can be shown using a **matrix of 0s and 1s**.

Suppose: $\mathbf{A} = \{a_1, a_2, \dots, a_m\}$ & $\mathbf{B} = \{b_1, b_2, \dots, b_n\}$

▶ The relation R can be represented by the matrix $\mathbf{M}_R = [m_{ij}]$, where

- **Rows** $\rightarrow$ elements of A
- **Columns** $\rightarrow$ elements of B

For each position (i, j) :

- **Put 1** if a_i is related to b_j
- **Put 0** if a_i is not related to b_j

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R, \\ 0 & \text{if } (a_i, b_j) \notin R. \end{cases}$$

- Consider the relation of who is enrolled in which class
 - Let $A = \{ \text{Alice, Bob, Claire, Dan} \}$
 - Let $B = \{ \text{CS101, CS201, CS202} \}$
 - $R = \{ (a,b) \mid \text{person } a \text{ is enrolled in course } b \}$

	CS101	CS201	CS202
Alice	X		
Bob		X	X
Claire			
Dan		X	X

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

EXAMPLE N

Let $A = \{1, 2, 3\}$ and $B = \{x, y\}$, Let R be a relation from A to B defined as $R = \{(1,y), (2,x), (2,y), (3,x)\}$

$$M = \begin{matrix} & \begin{matrix} x & y \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} \end{matrix}_{3 \times 2}$$

Examples of Representing Relations Using Matrices (*cont.*)

Example 3: Let $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3, b_4, b_5\}$. Which ordered pairs are in the relation R represented by the matrix

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix} ?$$

Solution:

$$R = \{(a_1, b_2), (a_2, b_1), (a_2, b_3), (a_2, b_4), (a_3, b_1), (a_3, b_3), (a_3, b_5)\}.$$

Matrices of Relations on Sets

If **R** is a **reflexive relation**, all the elements on the main diagonal of **M_R** are equal to 1.

$$\mathbf{M}_s = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrices of Relations on Sets

R is a symmetric relation

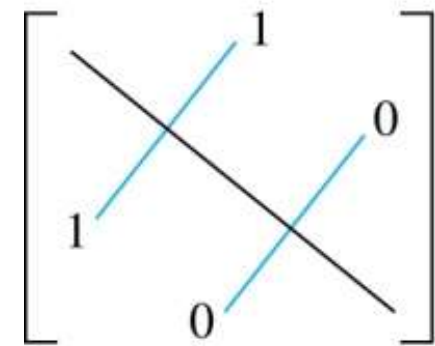
If a_i is related to a_j , then a_j is related to a_i .

In the matrix:

If position (i, j) is 1, then position (j, i) must also be 1.

So the matrix looks mirrored across the diagonal.

R is a symmetric relation,
if and only if $m_{ij} = 1$ whenever $m_{ji} = 1$.



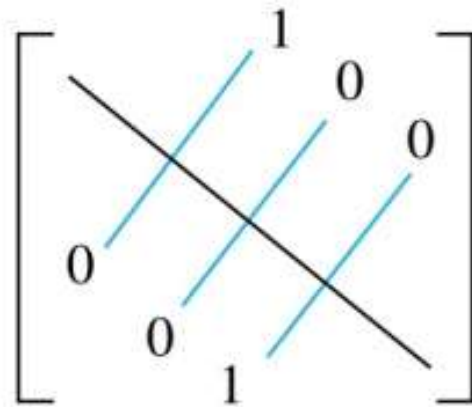
(a) Symmetric

Matrices of Relations on Sets

The **matrix of an antisymmetric relation** has the property that if $m_{ij}=1$ with $i \neq j$, then $m_{ji}=0$
OR Either $m_{ij}=0$ or $m_{ji}=0$ when $i \neq j$

In other words:

- If $i \neq j$, at least **one of the two entries must be 0.**



(b) Antisymmetric

Matrices of Relations on Sets

Suppose that the relation R on a set is represented by the matrix

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Is R reflexive, symmetric, and/or antisymmetric?

Solution:

Because all the diagonal elements are equal to 1, R is reflexive.

Because M_R is symmetric, R is symmetric and not antisymmetric because both $m_{1,2}$ and $m_{2,1}$ are 1.

REPRESENTING RELATIONS USING Directed Graph (Digraph)

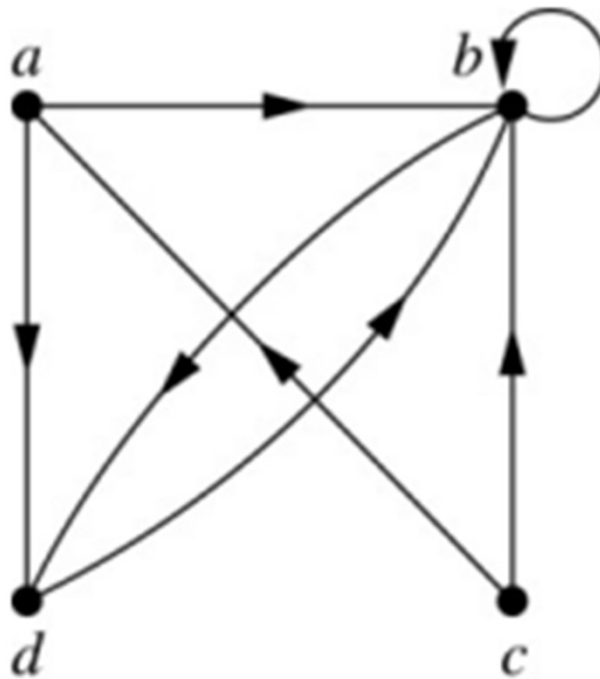
- A **directed graph** (or **digraph**) has:
 - A set of **vertices** (points or nodes) called **V**.
 - A set of **edges** (arrows) called **E**.
- Each edge is an **ordered pair** like (a, b) , meaning an arrow from **a** to **b**.
- The vertex **a** is the **start** (initial vertex) of the arrow.
- The vertex **b** is the **end** (terminal vertex) of the arrow.
- If an edge goes from a vertex back to itself, like (a, a) , it's called a **loop**.

In short:

- Vertices = points
- Edges = arrows between points
- Loop = arrow that starts and ends at the same point

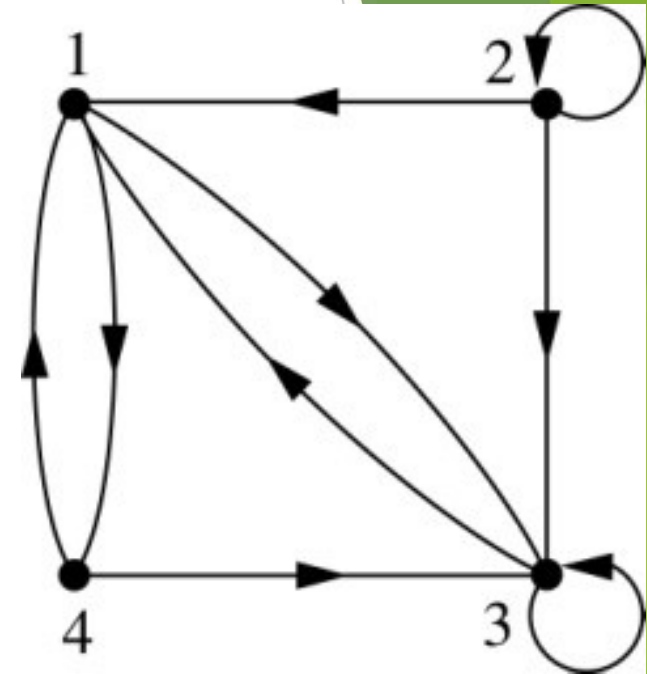
REPRESENTING RELATIONS USING Directed Graph (Digraph)

- A drawing of the directed graph with
- vertices a, b, c , and d , and
- edges $(a, b), (a, d), (b, b), (b, d), (c, a), (c, b), (d, b)$ is shown here.



REPRESENTING RELATIONS USING Directed Graph (Digraph)

Example : What are the ordered pairs in the relation represented by this directed graph?



Solution: The ordered pairs in the relation are $(1, 3)$, $(1, 4)$, $(2, 1)$, $(2, 2)$, $(2, 3)$, $(3, 1)$, $(3, 3)$, $(4, 1)$, and $(4, 3)$

EXAMPLE 0

For the relation matrix.

$$M = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{matrix}$$

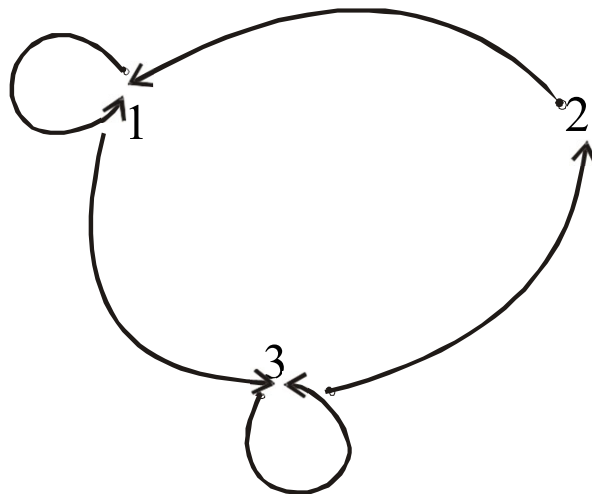
- List the set of ordered pairs represented by M.
- Draw the directed graph of the relation.

SOLUTION:

The relation corresponding to the given Matrix is

$$R = \{(1,1), (1,3), (2,1), (3,2), (3,3)\}$$

and its Directed graph is given below



Determining which Properties a Relation has from its Digraph

1. Reflexive: Every vertex must have a loop.

That means:

For every element x

The pair (x, x) must exist

In the graph: Every vertex must have a **small circle (loop)** on itself.

If even **one vertex** does not have a loop, Then it is **NOT reflexive**.

2. Symmetric

If there is an arrow from x to y ,
there must also be an arrow from y to x .

So If (x, y) exists, then (y, x) must also exist.

In the graph: Every arrow must have a matching arrow going back.

Example:

$x \rightarrow y$

$y \rightarrow x$

If one direction is missing, Then it is **NOT symmetric**.

3. Antisymmetric

- If $x \rightarrow y$ exists (and $x \neq y$),
then $y \rightarrow x$ must NOT exist.

So you cannot have arrows both ways between two different vertices.

If you see: $x \rightarrow y$

and $y \rightarrow x$

and $x \neq y$, then it is NOT antisymmetric.

4. Transitive: This means:

If you can go from $x \rightarrow y$

and from $y \rightarrow z$

then you must also be able to go from $x \rightarrow z$.

So if: $x \rightarrow y$

$y \rightarrow z$

Then $(x \rightarrow z)$ must also be present.

If that direct arrow is missing

Then it is NOT transitive.

DIRECTED GRAPH OF A REFLEXIVE RELATION

The directed graph of every reflexive relation includes an arrow from every point to the point itself (i.e., a loop).

EXAMPLE Q

Let $A = \{1, 2, 3, 4\}$ and define relations R_1 , R_2 , R_3 , and R_4 on A by

$$R_1 = \{(1, 1), (3, 3), (2, 2), (4, 4)\}$$

$$R_2 = \{(1, 1), (1, 4), (2, 2), (3, 3), (4, 3)\}$$

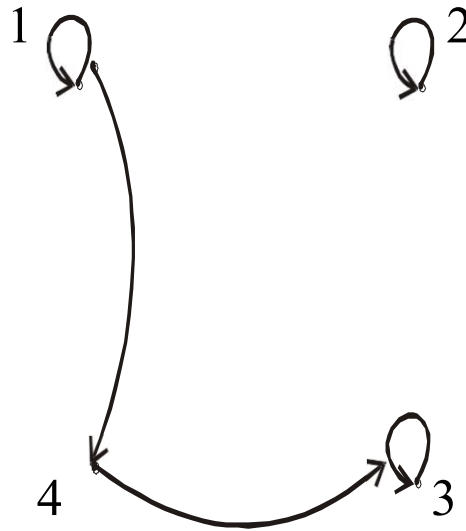
$$R_3 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$$

$$R_4 = \{(1, 4), (2, 2), (2, 3), (4, 1), (3, 3)\}$$

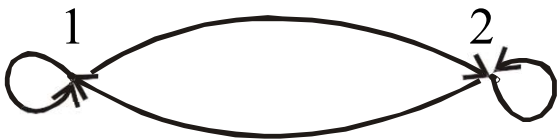
Then their directed graphs are:



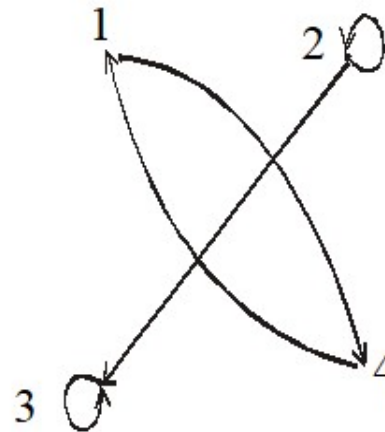
R_1 is reflexive because at every point of the set A we have a loop in the graph.



R_2 is not reflexive, as there is no loop at 4.



R_3 is reflexive



R_4 is not reflexive, as there are no loops at 1 and 4.

GRAPH OF A SYMMETRIC RELATION

For a symmetric directed graph whenever there is an arrow going from one point of the graph to a second, there is an arrow going from the second point back to the first.

EXAMPLE S

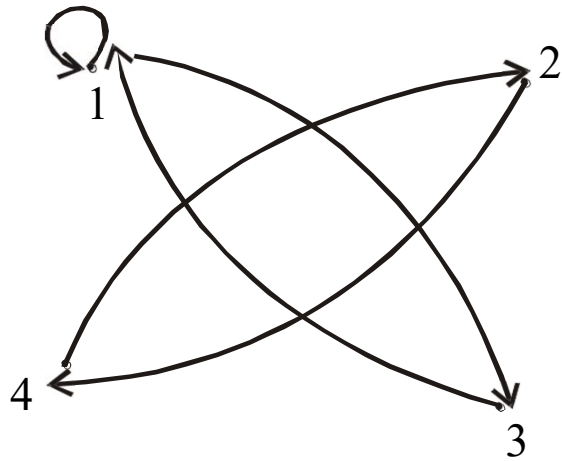
Let $A = \{1, 2, 3, 4\}$ and define relations R_1 , R_2 , R_3 , and R_4 on A by the directed graphs:

$$R_1 = \{(1, 1), (1, 3), (2, 4), (3, 1), (4, 2)\}$$

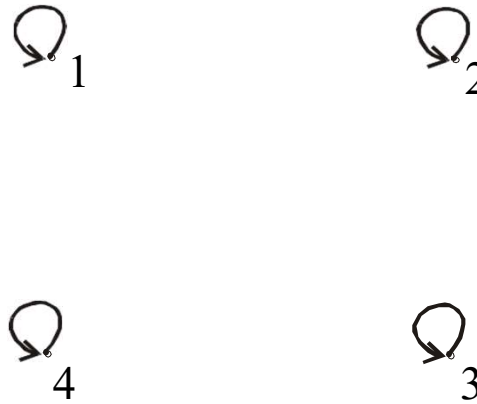
$$R_2 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$$

$$R_3 = \{(2, 2), (2, 3), (3, 4)\}$$

$$R_4 = \{(1, 1), (2, 2), (3, 3), (4, 3), (4, 4)\}$$

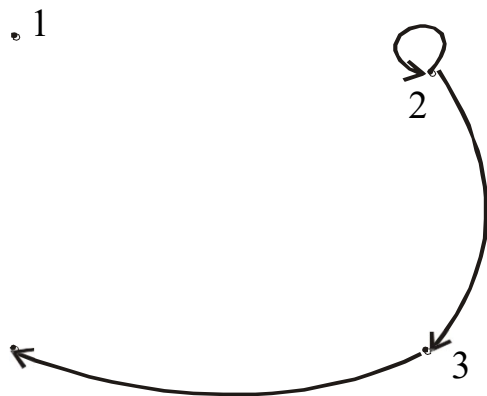


R_1 is symmetric



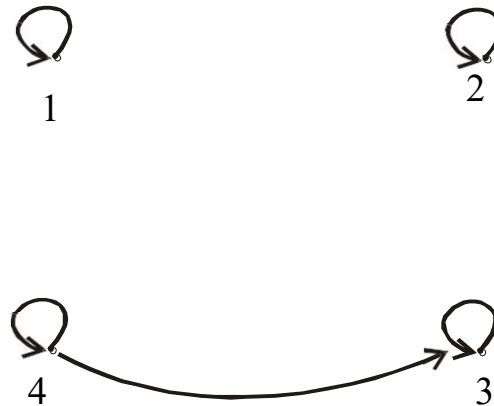
R_2 is symmetric

Because reversing any pair gives the same pair, which is already in the relation.



R_3 is not symmetric since there are arrows from 2 to 3 and from 3 to 4 but not conversely

But R_3 is antisymmetric.



R_4 is not symmetric since there is an arrow from 4 to 3 but no arrow from 3 to 4

But R_4 is antisymmetric.

DIRECTED GRAPH OF A TRANSITIVE RELATION

For a transitive directed graph, whenever there is an arrow going from one point to the second, and from the second to the third, there is an arrow going directly from the first to the third.

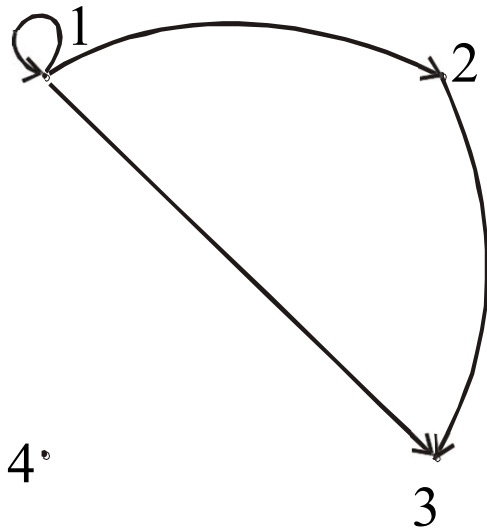
EXAMPLE W

Let $A = \{1, 2, 3, 4\}$ and define relations R_1 , R_2 and R_3 on A by the directed graphs:

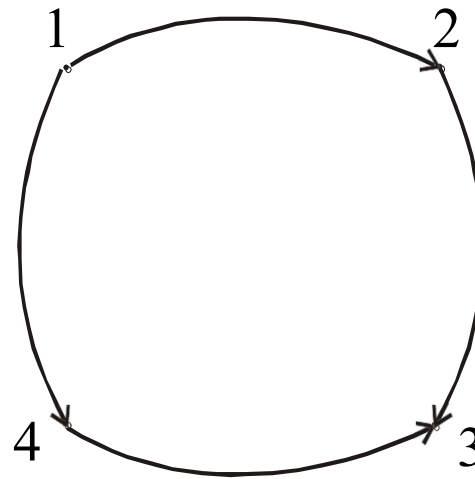
$$R_1 = \{(1, 1), (1, 2), (1, 3), (2, 3)\}$$

$$R_2 = \{(1, 2), (1, 4), (2, 3), (4, 3)\}$$

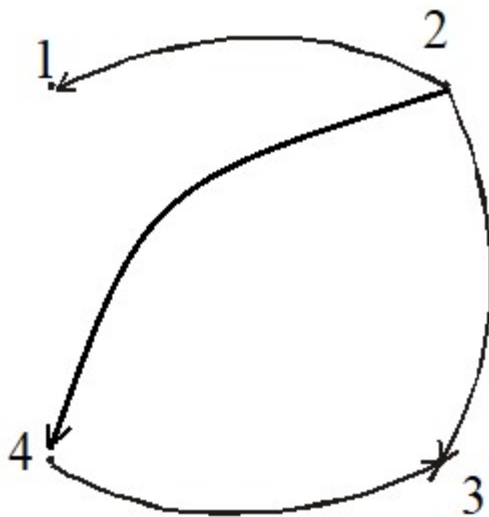
$$R_3 = \{(2, 1), (2, 4), (2, 3), (4, 3)\}$$



R_1 is transitive

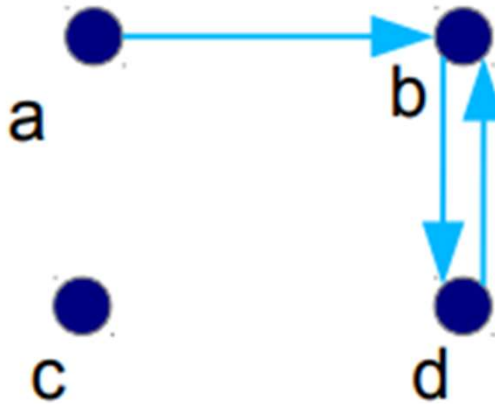


R_2 is not transitive since there is an arrow from 1 to 2 and from 2 to 3 but no arrow from 1 to 3 directly



R_3 is transitive

Determining which Properties a Relation has from its Digraph



- Reflexive? No, there are no loops
- Symmetric? No, there is an edge from a to b, but not from b to a
- Antisymmetric? No, there is an edge from d to b and b to d
- Transitive? No, there are edges from a to c and from c to b, but there is no edge from a to d